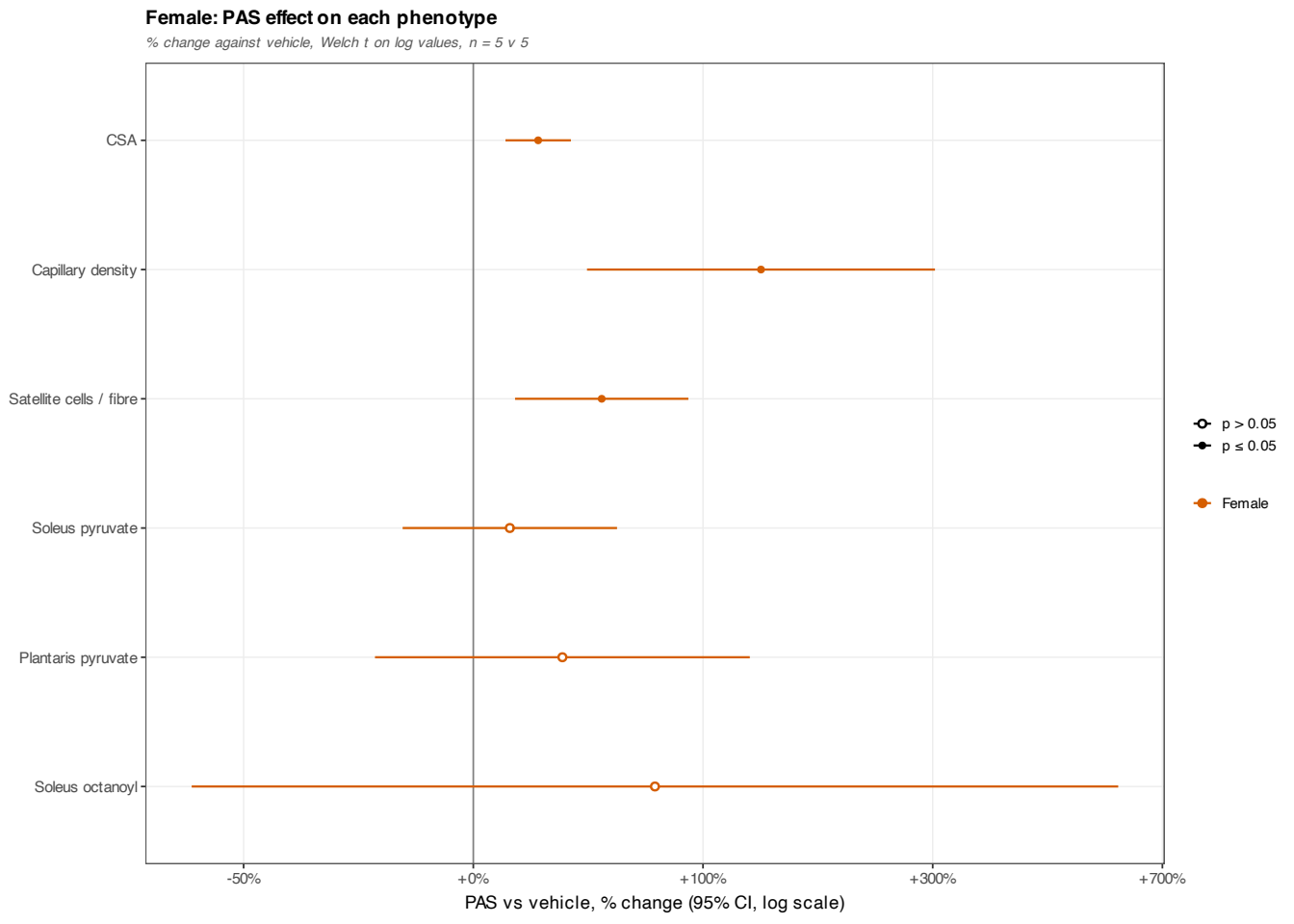
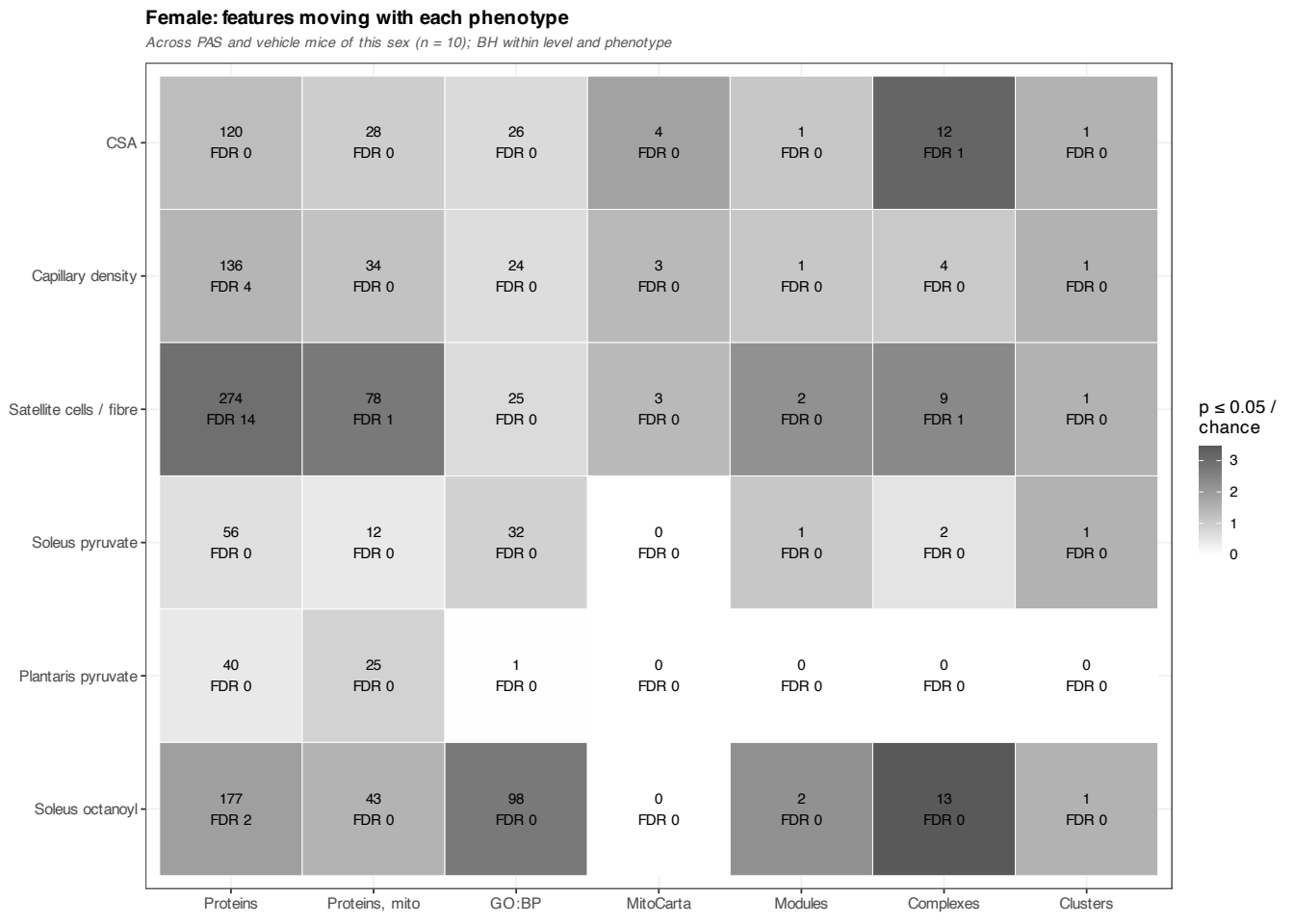
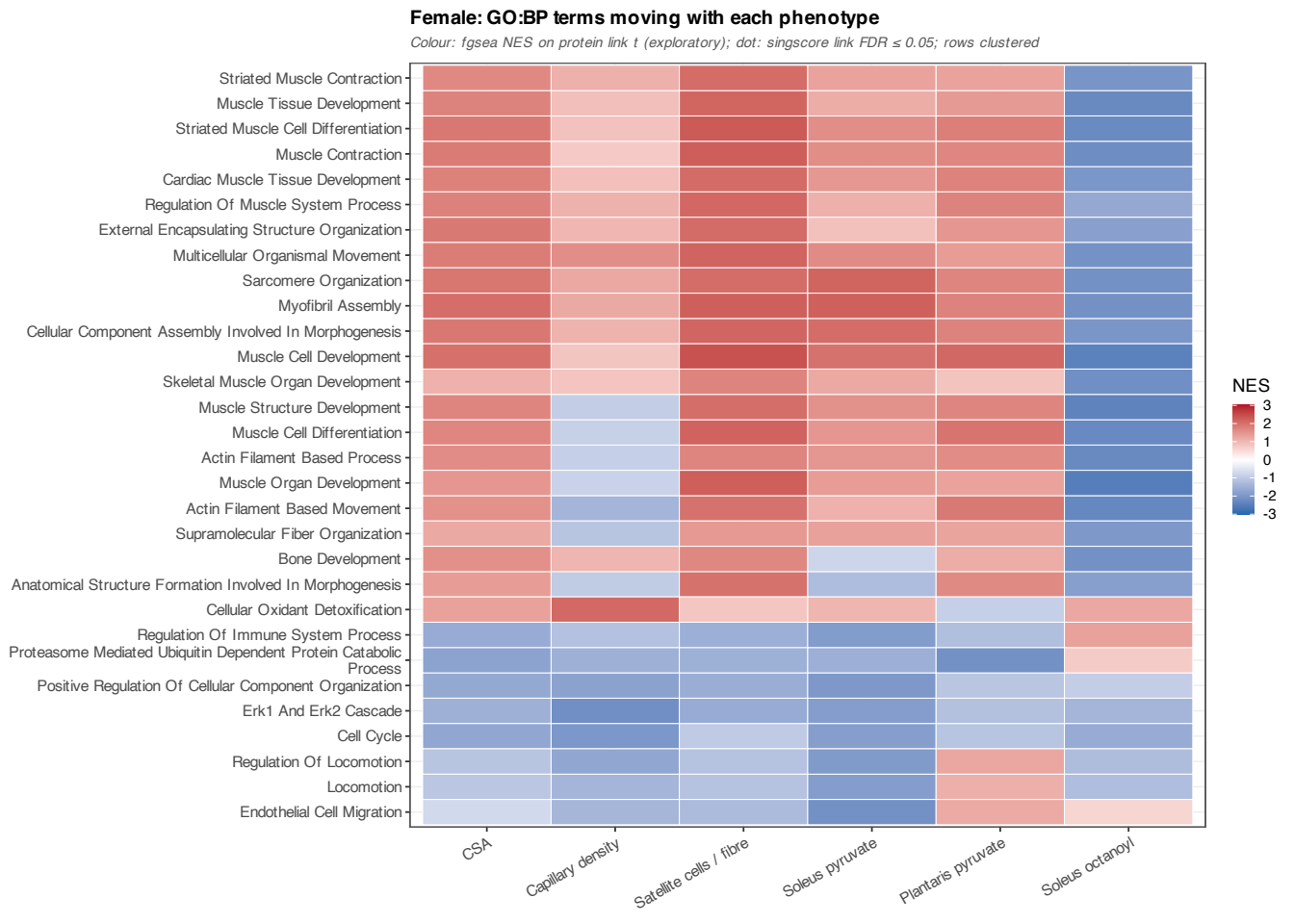
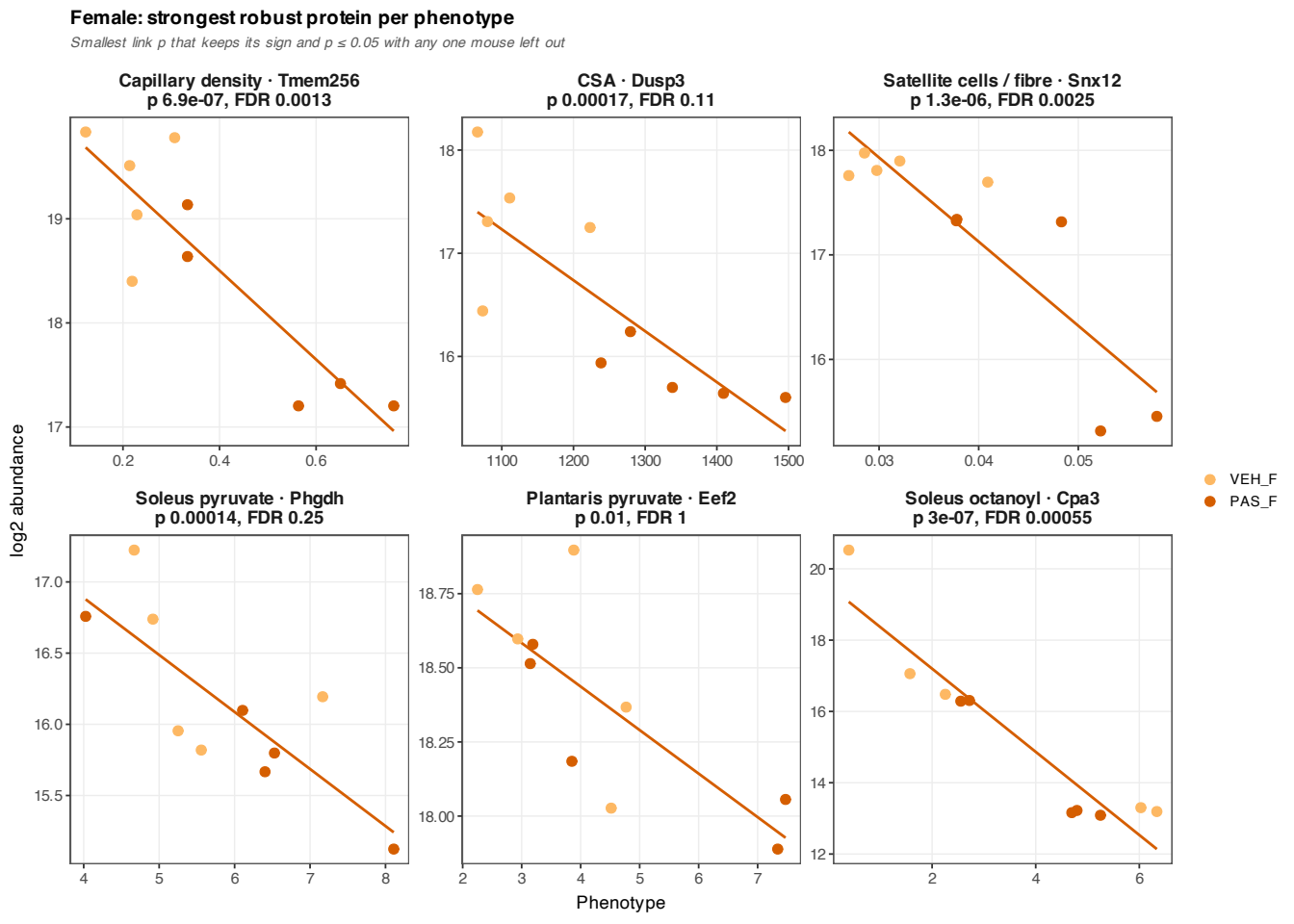

Female: PAS effect on each phenotype
% change against vehicle, Welch t on log values, n = 5 v 5
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
p > 0.05
p ≤ 0.05
Female
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)


Female: features moving with each phenotype
Across PAS and vehicle mice of this sex (n = 10); BH within level and phenotype
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
120 FDR 0
28 FDR 0
26 FDR 0
4 FDR 0
1 FDR 0
12 FDR 1
1 FDR 0
136 FDR 4
34 FDR 0
24 FDR 0
3 FDR 0
1 FDR 0
4 FDR 0
1 FDR 0
274 FDR 14
78 FDR 1
25 FDR 0
3 FDR 0
2 FDR 0
9 FDR 1
1 FDR 0
56 FDR 0
12 FDR 0
32 FDR 0
0 FDR 0
1 FDR 0
2 FDR 0
1 FDR 0
40 FDR 0
25 FDR 0
1 FDR 0
0 FDR 0
0 FDR 0
0 FDR 0
0 FDR 0
177 FDR 2
43 FDR 0
98 FDR 0
0 FDR 0
2 FDR 0
13 FDR 0
1 FDR 0
Proteins
Proteins, mito
GO:BP
MitoCarta
Modules
Complexes
Clusters
p ≤ 0.05 / chance
3
2
1
0


Female: GO:BP terms moving with each phenotype
Colour: fgsea NES on protein link t (exploratory); dot: singscore link FDR ≤ 0.05; rows clustered
Striated Muscle Contraction
Muscle Tissue Development
Striated Muscle Cell Differentiation
Muscle Contraction
Cardiac Muscle Tissue Development
Regulation Of Muscle System Process
External Encapsulating Structure Organization
Multicellular Organismal Movement
Sarcomere Organization
Myofibril Assembly
Cellular Component Assembly Involved In Morphogenesis
Muscle Cell Development
Skeletal Muscle Organ Development
Muscle Structure Development
Muscle Cell Differentiation
Actin Filament Based Process
Muscle Organ Development
Actin Filament Based Movement
Supramolecular Fiber Organization
Bone Development
Anatomical Structure Formation Involved In Morphogenesis
Cellular Oxidant Detoxification
Regulation Of Immune System Process
Proteasome Mediated Ubiquitin Dependent Protein Catabolic Process
Positive Regulation Of Cellular Component Organization
Erk1 And Erk2 Cascade
Cell Cycle
Regulation Of Locomotion
Locomotion
Endothelial Cell Migration
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
NES
3
2
1
0
-1
-2
-3


Female: strongest robust protein per phenotype
Smallest link p that keeps its sign and p ≤ 0.05 with any one mouse left out
Capillary density · Tmem256
p 6.9e-07, FDR 0.0013
CSA · Dusp3
p 0.00017, FDR 0.11
Satellite cells / fibre · Snx12
p 1.3e-06, FDR 0.0025
Soleus pyruvate · Phgdh
p 0.00014, FDR 0.25
Plantaris pyruvate · Eef2
p 0.01, FDR 1
Soleus octanoyl · Cpa3
p 3e-07, FDR 0.00055
log2 abundance
Phenotype
VEH_F
PAS_F